



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin

Ollscoil Átha Cliath | The University of Dublin

Faculty of Engineering, Mathematics and Science

School of Computer Science & Statistics

Integrated Computer Science Programme
Year 4 Annual Examinations

Semester 1 2019

Next Generation Networks

Wednesday 11 December

RDS SIM COURT

17.00-19.00

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Instructions to Candidates:

- **Answer FOUR questions from Part A, and EIGHT questions from Part B.**

Materials Permitted for this Examination:

- **Calculator**
- **Mathematical Tables**

PART A

Q.1

With regards to wireless local area networks (WLAN):

a) Describe the motivation behind their deployment.

[6 marks]

b) Describe their limitations.

[6 marks]

c) Outline the access point functionalities and mention the main applications of the infrastructure-based mode.

[6 marks]

Q.2

With regards to cognitive radio:

a) What are its desirable features?

[6 marks]

b) What are its applications?

[6 marks]

c) Describe its cross layer issues.

[6 marks]

Q.3

In a certain system the arrivals, which can be modelled as a Poisson process, occur at a rate of one every 200 ms, and the service times are exponentially distributed with mean 2 s. Also, there is no limit in the number of servers available.

a) Give the Kendall's notation, motivating your choice.

[6 marks]

b) Calculate the average number of customers in the system.

[6 marks]

c) Calculate the average delay per customer.

[3 marks]

d) Calculate the average queuing delay. What do you notice? Why it is so?

[3 marks]

Q.4

Considering an optical transmission systems for telecommunications networks:

- a) List three different types of optical fibre, explaining their main characteristics, indicating their main advantages and disadvantages.

[9 marks]

- b) Describe the role of optical amplifiers in a transmission system, how they are used, how they affect the noise in the system, what is the effect of different span distances and their main advantage and disadvantage over regenerators.

[9 marks]

Q.5

- a) Describe what are the main reasons behind the need for convergence of optical and wireless networks and why they are specifically needed for 5G networks. Then explain, using any appropriate diagram, what is the problem for transporting fronthaul signals over a Passive Optical Network (PON) and how can this be solved.

[9 marks]

- b) Describe all the evolutionary steps in fronthaul transmission, that brought base stations from being distributed (Distributed Radio Access Network -D-RAN) to becoming centralised (C-RAN), explaining the rationale behind any of these steps.

[9 marks]

Q.6

- a) An optical signal with bit rate of 25Gb/s runs over a single mode fibre with dispersion coefficient of 5ps/km/nm. The signal uses non-return-to-zero (NRZ) coding, with a bandwidth of 50 GHz. What is the maximum distance for the link if the maximum tolerable broadening of the optical pulses is equal to 60% of the bit time duration? (Assume the following conversion ratio: 100 GHz = 0.8 nm).

[9 marks]

- b) You need to achieve a link distance of 400 km and you must compensate dispersion using dispersion compensating fibre (DCF) with coefficient of -150ps/km/nm. Assuming that the system uses 4 intermediated amplification stations and that the DCF fibre is located in each of these stations, what length of DCF do you need at each station?

[9 marks]**PART B**

For each question you pick, select only ONE answer out of the four available options.

Q.7

With respects to optical fibres:

- a) Single mode fibres have a larger core than multi-mode fibres
- b) Single mode fibres typically use a step index between core and cladding
- c) Single mode fibre can transmit one wavelength channel at a time
- d) Single mode fibres can transmit only LP₀₁, LP₁₁ and LP₂₁ modes.

[3.5 marks]**Q.8**

Which one among the following options, **is not** a functionality of the Physical (PHY) layer in a wireless system?

- a) Frequency selection
- b) Error control
- c) Modulation
- d) Signal detection

[3.5 marks]

Q.9

Which of the following statements is **false** with respect to Passive Optical Networks (PONs):

- a) PONs are typically based on time division multiplexing
- b) PONs can operate more than one wavelength at a time
- c) PONs were developed to reduce the amount of electronic equipment required
- d) PONs use a continuous frame transmission in the upstream direction

[3.5 marks]

Q.10

Which one among the following options, **is not** a Long Term Evolution (LTE) performance objective?

- a) 100 Mbps peak data rate in downlink
- b) Reduced latency down to 10 ms
- c) Scalable bandwidth up to 20 MHz
- d) Mobility up to 700 km/h

[3.5 marks]

Q.11

Which of the following **is not** a reason why other technologies had to be developed in addition to Local Loop Unbundling (LLU)?

- a) LLU cannot offer differentiated quality of service to different operators
- b) LLU is expensive for other operators that need to bring equipment to all central offices where they serve customers
- c) LLU makes crosstalk difficult to tackle as vectoring cannot operate properly
- d) LLU does not allow other operators to collect aggregate customer traffic at few regional points of presence

[3.5 marks]

Q.12

In the a/b/m/K queuing theory notation, the symbol 'b' stands for:

- a) Type of arrival process
- b) Number of servers
- c) Maximum number of customers allowed in the system
- d) Service time distribution

[3.5 marks]

Q.13

Which of the following sentences about base station functional split **is false**?

- a) Split 8 is the one that requires the highest data rate
- b) Stack processing blocks (e.g., MAC, PHY, etc.) can only be split at their interfacing points and not within a block itself
- c) If the split selected is above the MAC then the latency requirements are more relaxed
- d) Functional split will also be used for 5G base stations

[3.5 marks]

Q.14

Which one among the following equations, correctly expresses the Little's law?

- a) $\lambda E[n] = E[\tau]$
- b) $E[n] = E[\tau]$
- c) $E[n] = \lambda^2 E[\tau]$
- d) None of the above is correct

[3.5 marks]

Q.15

What is the main benefit of introducing MPLS in a network?

- a) Faster packet switching due to the smaller address space
- b) It uses labels that have network-wide significance
- c) It enables tunnelling, which makes network management easier
- d) It uses Forward Error Correction

[3.5 marks]

Q.16

Which one among the following equations, correctly expresses the average delay for an M/M/1 queue?

- a) $E[\tau] = 1 / (\lambda - \mu)$
- b) $E[n] = 1 / (\mu - \lambda)$
- c) $E[\tau] = 1 / (\mu - \lambda)$
- d) None of the above is correct

[3.5 marks]

Q.17

Which of the following statements about Quality of Service **is false**?

- a) Strict priority does not allow packets from lower priority to be scheduled until the higher priority queues are empty
- b) Random Early Discard was developed taking into account how the TCP protocol manages congestion control
- c) Priority-based deficit weighted round robin can effectively increase the capacity of a channel
- d) A shaper operates by delaying some of the incoming packets

[3.5 marks]

Q.18

What is the maximum rate that the 802.11ay standard aims to achieve?

- a) In excess of 30 Gbps
- b) At most 10 Gbps
- c) At most 1 Gbps
- d) In excess of 60 Gbps

[3.5 marks]

Q.19

Which of the following **is not** an advantage of Software Defined Networking?

- a) It requires switches to run their routing algorithms in a distributed fashion
- b) It enables decoupling of control and data plane
- c) It allows an external entity to remove elements from a switch flow table
- d) It can operate on more than one layer of the protocol stack

[3.5 marks]

Q.20

Which one among the following options, **cannot** be considered a cognitive radio meter?

- a) Occupied bands
- b) Signal strength
- c) Neighbour list
- d) Waveform

[3.5 marks]

Q.21

A reconfigurable Optical Add Drop Multiplexer (ROADM):

- a) Is a system that can incorporate Wavelength Selective Switches
- b) Was introduced in the first generation of optical networks
- c) Is the core element in an IP-over-WDM network
- d) Can route data based on either Ethernet or IP destination addresses

[3.5 marks]

Q.22

How can we cope with the demand for higher capacity?

- a) Improving the radio link efficiency
- b) Getting more spectrum
- c) Deploying more nodes in the wireless infrastructure
- d) By using all the above strategies

[3.5 marks]